

Hackthon Project Phases Template

Project Title:

Audio Transcription app using open Ai Whisper

Team Name:

CODE BREAKERS

Team Members:

- Vengala Akhilan
 - Shetty Sai Kumar
 - Thanda Tharun Teja
 - Palle Jayanth
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Phase-1: Brainstorming & Ideation

Objective:

Develop an AI-powered transcription tool to efficiently convert audio into text.

Key Points:

1. Problem Statement:

- Users face difficulty in manually transcribing long audio content, which is tedious and prone to errors.
- Many current transcription tools struggle with accuracy, especially with accents, background noise, or multiple speakers.

2. Proposed Solution:

- Leverage Open AI's Whisper to provide real-time, accurate transcriptions with support for various languages and accents.
- Automatically identify and differentiate speakers in conversations, and include timestamps for easy reference in long transcriptions.
- Translate text from one language to another targeted language selected by the user.
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3. Target Users:

- Those who need transcriptions of lectures, seminars, or interviews for study and research purposes.
- People who need quick transcriptions of meetings, podcasts, or video content for documentation and accessibility.

4. Expected Outcome:

- Provide fast, reliable transcriptions with minimal manual effort, saving time and to increase productivity.
- Enable users with hearing impairments or non-native speakers to easily access and understand audio-based content.

Phase-2: Requirement Analysis

Objective:

Define the technical and functional requirements for the transcription app.

Key Points:

1. Technical Requirements:

- **Programming Language:** Python will be used for backend development and API integration.
- **Backend:** Google Gemini Flash API for real-time vehicle data fetching.
- **Frontend:** Frontend: React.js & Stream lit Web Framework for interactive and user-friendly UI.
- **Database:** No database required initially; API-based queries for data retrieval.

2. Functional Requirements:

- User Management & Authentication
- Audio File Upload & Management
- Accurate Audio Transcription with Multi-language Support

- Text Editing & Formatting
- Export Options (TXT, DOCX, PDF, SRT)
- Cloud Storage Integration

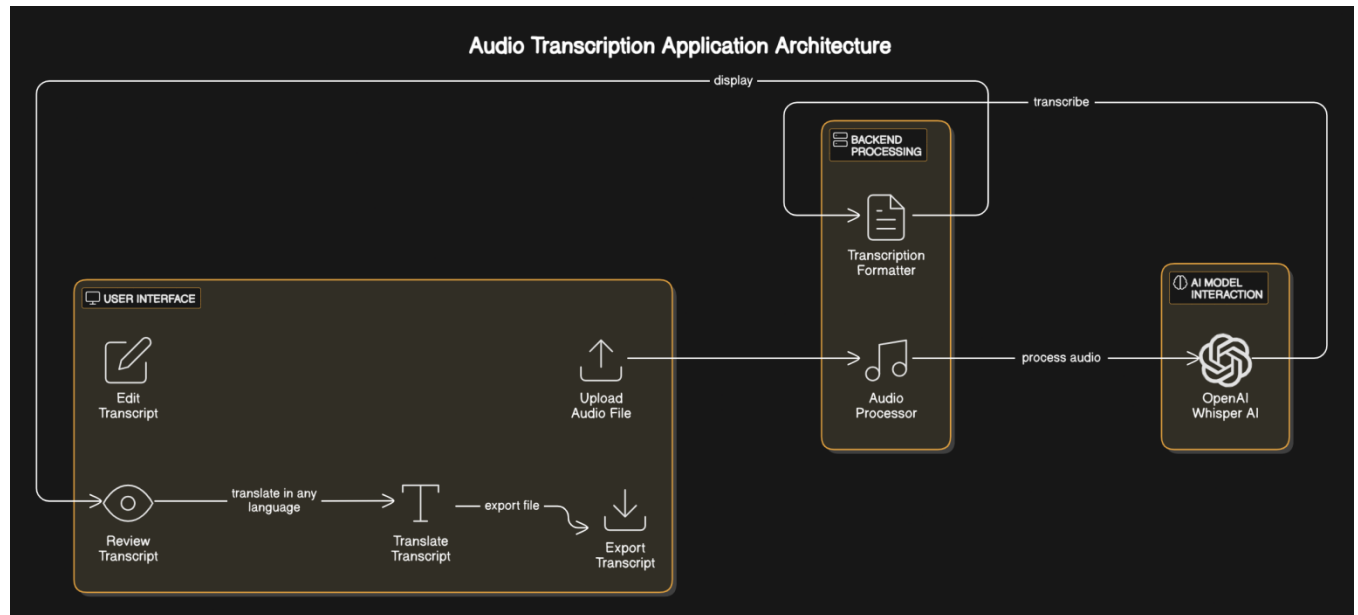
3. Constraints & Challenges:

- Ensuring quick and accurate real-time data from Whisper API.
- Efficiently managing API calls to stay within rate limits.
- Ensuring transcription accuracy for various languages and accents.
- Managing background noise and multiple speakers.
- Optimizing processing speed for large audio files.

Phase-3:Project Design

Objective:

Develop the system architecture and user flow



Key Points:

1. System Architecture:

- Users upload audio files.
- Backend processes audio using Open AI Whisper.
- AI model transcribes, formats, and applies timestamps.
- Users review, edit, and export transcripts.

2. User Flow:

- User uploads an audio file (MP3/WAV).
- The app processes the file using Whisper AI.
- Transcription is generated and displayed.
- User edits and exports the transcript.

3. UI/UX Considerations:

- Minimalist, user-friendly interface.
 - Dark & Light mode for accessibility.
 - Real-time text formatting & editing
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Phase-4:Project Planning (Agile Methodologies)

Objective:

Break down development tasks for efficient completion.

Sprint	Task	Priority	Duration	Deadline	Assigned To	Dependencies	Expected Outcome
Sprint 1	EnvironmentSetup & API Integration	● High	6 hours (Day 1)	EndofDay 1	Akhilan	Google API Key, Python,google collab	APIconnection established &working
Sprint 1	Frontend UI Development	● Medium	4 hours (Day 1)	EndofDay 1	Akhilan	API response format finalized	BasicUIwithinput fields
Sprint 2	Transcription & Formatting	● High	5 hours (Day 1)	EndofDay 1	Tharun Teja	API response,UI elements ready	Searchfunctionality with filters
Sprint 2	ErrorHandling ,Debugging	● High	5 hours (Day 1)	EndofDay 1	Sai Kumar	API logs,UI inputs	ImprovedAPI stability
Sprint 3	Testing & UI Enhancements	● Medium	5 hours (Day 2)	Mid-Day 2	Jayanth	APIresponse,UI layout completed	ResponsiveUI, better user experience
Sprint 3	FinalPresentation & Deployment	● Low	2 hour (Day 2)	EndofDay 2	Entire Team	Working prototype	Demo-ready project

(● HighPriority)

(● HighPriority)

(● MediumPriority)

Phase-5: Project Development

Objective:

Implement core features of the transcription app.

Key Points:

1. Technology Stack Used:

- Frontend: Streamlit
- Backend: OpenAI Whisper
- Database: no database used.
- File Processing: FFmpeg
- Cloud Storage: no need.

2. Development Process:

- Implement API key authentication and Whisper AI integration.

- Develop real-time transcription and text formatting features.
 - Optimize processing for large audio files.
3. **Challenges & Fixes:**
- **Challenge:** Handling accents and background noise.
 - **Fix:** Use advanced noise reduction and context-aware AI models.
 - **Challenge:** Large file processing speed.
 - **Fix:** Implement efficient chunk processing.

Phase-6: Functional & Performance Testing

Objective:

Ensure the app performs accurately and efficiently.

Test Case ID	Category	Test Scenario	Expected Outcome	Status	Tester
TC-001	Functional Testing	Transcribe an English audio file	Accurate transcription of the audio in text format	✔ Passed	Tester 1
TC-002	Functional Testing	Transcribe and translate a Spanish audio file into English	Correct transcription and translation output	✔ Passed	Tester 2
TC-003	Performance Testing	Process a 30-minute audio file	API should return results within an acceptable time frame	⚠ Needs Optimization	Tester3
TC-004	Bug Fixes & Improvements	Fixed incorrect word recognition in noisy audio	Improved transcription accuracy	✔ Fixed	Developer
TC-005	Final Validation	Check UI responsiveness for file uploads and results display	UI should function properly on both mobile and desktop	✔ Passed	Tester2
TC-006	Deployment Testing	Deploy the app using Streamlit Sharing	App should be accessible online	🚀 Deployed	DevOps

Final Deliverables

1. **Project Report based on the templates**
2. **Demo Video (3-5 Minutes)**
3. **Git Hub /Code Repository Link**
4. **Presentation**